EF5560: FinTech and AI in Finance

Course Syllabus, Semester A 2026/27

City University of Hong Kong, Department of Economics and Finance

Instructor	Prof. Gavin Feng
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Credit units	3
Medium	English
Prerequisites	None. See *Course purpose* for what Classes 1–5 assume
Class format	Twelve three-hour seminars; exact dates, time, and venue are announced through the official timetable and Canvas
Course site	Official announcements, course materials, submission links, and assessment instructions will be posted on Canvas
Consultation	Zoom office hours at 10:00 on the Sundays listed on Canvas, where the link is also posted Teaching assistants: Yunting Liu, [contact removed] Anna Zhao, [contact removed] Copy me on any email you send them, with the same `[EF5560]` subject prefix

Course purpose

This course has two parts. In Classes 1–5, students use AI-assisted coding in R and Python to obtain financial data, construct predictors, study return predictability, compare linear and nonlinear models, interpret model outputs, and turn forecasts into portfolio decisions. You will not be asked to memorize code. You will be asked to define the problem, check when each variable became available, protect the validation and test samples, and report weak results as weak.

Classes 6–12 study how FinTech businesses work: the technology they use, how they make money, the risks they create, and how they are regulated. We cover blockchain, Bitcoin and Ethereum, decentralized finance, tokenization, stablecoins and central bank digital currencies, payment systems, lending, robo-advising, cybersecurity, RegTech, and selected AI applications for financial and compliance text.

Classes 1–5 build on each other, so regular attendance is important.

Learning outcomes

The course aims to help students:

1. use selected AI-assisted R and Python workflows to organize, visualize, and check the timing of financial data;
2. interpret and compare selected regression and machine-learning methods for return prediction and portfolio applications using out-of-sample evidence, including the limits of coefficients, feature importance, and response profiles;
3. explain core mechanisms and business models in selected FinTech areas and recognize major financial, operational, cybersecurity, conduct, and regulatory risks; and
4. communicate an evidence-based analysis of a FinTech company or financial machine-learning application.

Coverage and depth depend on class progress. Students are assessed only on material taught in class or released through Canvas.

Assessment

Assessment component	Weight
FinTech company case report	10%
Recorded video presentation	20%
Machine-learning investment assignment	20%
Participation and in-class work	10%
Two-hour final examination	40%
Total continuous assessment	60%
Final examination	40%

FinTech company case project. Students use the same teams as for the machine-learning investment assignment (up to three students). Each team submits a written case report (10%, no more than ten pages) and a Zoom recording link (20%, no more than ten minutes) by 12:00 noon on 29 November 2026. Every member must deliver at least two minutes of substantive presentation. Include the Zoom recording link on the report's first page. Do not set a password for the recording. There is no in-person presentation. A separate brief gives the framework, evidence standards, and rubric.

Machine-learning investment assignment. Teams of up to three submit one joint report, portfolio-weight CSV, and reproducible code package. It is released after Class 4 and is due on 15 October 2026 at 2:00 p.m. The report cover lists all team members and CityU student numbers; one member is the designated submitter. The assignment brief gives the required analysis, submission instructions, and grading criteria.

Teaching-assistant tutorials. Two Sunday office-hour slots are used for TA tutorials: 27 September 2026 on preparing the machine-learning assignment, and 6 December 2026 to work through one practice exam. Both are recorded.

Participation. Everyone starts at five marks out of ten and the mark moves during the semester.

- **Cold call.** I call on students by name rather than wait for volunteers, so expect to be asked.
- **A correct or well-reasoned answer earns one mark.** Sound reasoning counts even when the final answer is not the expected one.
- **A wrong answer costs nothing.** Being wrong, or passing, leaves the mark unchanged.
- **Being absent when called costs one mark.** Email me at least a day ahead if you cannot attend your registered class; otherwise the mark falls by one.

Questions, short exercises, and useful feedback to classmates count toward the same ten marks.

Final examination. The examination covers both course blocks and tests concepts, mechanisms, calculations, and critical comparison. It is handwritten and held during the examination period. Practice exams are released on Canvas in mid-November; the second teaching-assistant tutorial works through one of them. Permitted materials will be confirmed on Canvas, subject to the final University examination instructions.

Tentative teaching schedule

The order and emphasis may be adjusted according to class progress and current developments.

Class	Block	Topic and focus
1	ML in Finance	Financial data and vibe coding in R and Python
2	ML in Finance	Regression, return predictability, and out-of-sample design

Class	Block	Topic and focus
3	ML in Finance	Linear machine learning for return prediction
4	ML in Finance	Nonlinear machine learning for return prediction
5	ML in Finance	Interpretable machine learning and portfolio decisions
6	FinTech	FinTech overview and blockchain
7	FinTech	Bitcoin and Ethereum
8	FinTech	DeFi, NFTs, and real-world assets
9	FinTech	Stablecoins and CBDCs
10	FinTech	Payment systems
11	FinTech	Lending and crowdfunding
12	FinTech	Robo-advising, cybersecurity, RegTech, and AI applications to financial text, with a closing course synthesis

Course practices and policies

- **Evidence.** Time-sensitive claims about markets, companies, and regulation must be supported by a dated and credible source.
- **Reproducibility.** Empirical work must preserve the data timing and transformation steps needed to reproduce the reported result.
- **Generative AI.** AI-assisted coding is used in Classes 1–5. AI may be used in coursework when permitted by the relevant instructions and disclosed. Students remain responsible for all submitted work. Generative-AI tools are not permitted in the final examination.
- **Academic integrity.** Students must submit their own work, acknowledge others' words, graphics, ideas, and data, and report findings honestly. Plagiarism, fabrication, undisclosed copying, and submission of unverified generated material are serious offences under the University's Rules on Academic Honesty. All new students admitted in Semester A 2026/27 must complete the online tutorial and quiz and submit the declaration through the Academic Honesty website no later than **30 November 2026**.
- **Announcements.** Later Canvas or University announcements take precedence.

Official course information

See the 2026/27 University course catalogue.